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Casing assembly and electronic device

Abstract

The casing assembly includes a casing, a cage, a supporting component, an engagement component and an operation assembly. The cage is movably disposed in the casing. One end of the supporting component is pivotably disposed on the casing, and another end of the supporting component is slidably disposed on the cage. The engagement component is pivotably disposed on the cage and selectively engaged with the supporting component to position the cage. The operation assembly is movably disposed on the cage and connected to the engagement component. The operation assembly is movable relative to the cage, and the engagement component is driven to disengage from the supporting component via the operation assembly selectively.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This non-provisional application claims priority under 35 U.S.C. § 119(a) on Patent Application No(s). 202310828698.5 filed in China on Jul. 6, 2023, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

(2) The disclosure relates to a casing assembly and an electronic device.

BACKGROUND

(3) An electronic device such as a server is usually equipped with an electronic module, such as a hard disk drive module (hereinafter “HDD module”). The cage of the HDD module is pivotably disposed on a server casing. After the cage is pivoted relative to the server casing, an installation opening of the cage can be shifted from its original position hidden inside the server casing to being exposed on the outside, so that a user or a maintainer can remove hard disk drives from the cage or insert the hard disk drives into the cage through the installation opening.

(4) Conventionally, the cage is connected to a handle via a linkage mechanism, and the rotation movement of the cage is driven by the handle and the linkage mechanism. After the cage is pivoted relative to the server casing, the user or the maintainer needs to hold the handle for maintaining the cage at a position where the installation opening is exposed outside the server casing. However, the handle is exposed outside the server for being operated by the use or the maintainer, which adversely affects the appearance of the server. In addition, the structure of the link mechanism connecting the handle and the cage is too complicated. Accordingly, how to solve the aforementioned issues is a crucial topic in this field.

SUMMARY

(5) The disclosure provides a casing assembly and an electronic device, which enable the cage to

be fixed in the position where the installation opening thereof is exposed outside the server casing without the need for a handle while maintaining the appearance of the electronic device.

(6) One embodiment of the disclosure provides a casing assembly. The casing assembly includes a casing, a cage, a supporting component, an engagement component and an operation assembly. The cage is movably disposed in the casing. One end of the supporting component is pivotably disposed on the casing, and another end of the supporting component is slidably disposed on the cage. The engagement component is pivotably disposed on the cage and selectively engaged with the supporting component to position the cage. The operation assembly is movably disposed on the cage and connected to the engagement component. The operation assembly is movable relative to the cage, and the engagement component is driven to disengage from the supporting component via the operation assembly selectively.

(7) Another embodiment of the disclosure provides an electronic device. The electronic device includes a casing assembly. The casing assembly includes a casing, a cage, a supporting component, an engagement component and an operation assembly. The cage is movably disposed in the casing. One end of the supporting component is pivotably disposed on the casing, and another end of the supporting component is slidably disposed on the cage. The engagement component is pivotably disposed on the cage and selectively engaged with the supporting component to position the cage. The operation assembly is movably disposed on the cage and connected to the engagement component. The operation assembly is movable relative to the cage, and the engagement component is driven to disengage from the supporting component via the operation assembly selectively.

(8) According to the casing assembly and the electronic devices as discussed in the above embodiments, one end of the supporting component is pivotably disposed on the casing, and another end of the supporting component is slidably disposed on the cage, the engagement component is pivotably disposed on the cage and is selectively engaged with the supporting component to position the cage, and the operation assembly is movably disposed on the cage and can drive the engagement component to disengage from the supporting component, and thus in a case that there is no handle provided in the casing assembly, the cage can still be fixed in the opened position or released from the opened position, while the appearance of the electronic device can be maintained.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The present disclosure will become better understood from the detailed description given herein below and the accompanying drawings which are given by way of illustration only and thus are not intending to limit the present disclosure and wherein:
- (2) FIG. 1 is a partially perspective view of an electronic device according to a first embodiment of the disclosure;
- (3) FIG. 2 is a partially exploded view of the electronic device in FIG. 1;
- (4) FIG. 3 is another partially exploded view of the electronic device in FIG. 1;
- (5) FIG. 4 is a partially cross-sectional view of the electronic device in FIG. 1;
- (6) FIG. 5 is a partially cross-sectional view of the electronic device in FIG. 4 when a slide knob is operated;
- (7) FIG. 6 is a partially cross-sectional view of the electronic device in FIG. 5 when a torsion spring lifts up a cage;
- (8) FIG. 7 is a partially cross-sectional view of the electronic device in FIG. 6 when the cage is pivoted relative to a casing;
- (9) FIG. 8 is a partially cross-sectional view of the electronic device in FIG. 7 when an engagement

component is engaged with a supporting component;

(10) FIG. 9 is a partially perspective view of the electronic device in FIG. 1 when the casing is in an opened position;

(11) FIG. 10 is a partially cross-sectional view of the electronic device in FIG. 8 when the engagement component is disengaged from the supporting component;

(12) FIG. 11 is a partially cross-sectional view of an electronic device according to a second embodiment of the disclosure;

(13) FIG. 12 is a partially cross-sectional view of the electronic device in FIG. 11 when a button is pressed;

(14) FIG. 13 is a partially cross-sectional view of the electronic device in FIG. 11 when an engagement component is engaged with a supporting component;

(15) FIG. 14 is a partially cross-sectional view of the electronic device in FIG. 13 when the engagement component is disengaged from the supporting component;

(16) FIG. 15 is a partially cross-sectional view of an electronic device according to a third embodiment of the disclosure;

(17) FIG. 16 is a partially cross-sectional view of the electronic device in FIG. 15 when a slide knob is operated;

(18) FIG. 17 is a partially cross-sectional view of the electronic device in FIG. 15 when an engagement component is engaged with a supporting component; and

(19) FIG. 18 is a partially cross-sectional view of the electronic device in FIG. 17 when the engagement component is disengaged from the supporting component.

DETAILED DESCRIPTION

(20) In the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. It will be apparent, however, that one or more embodiments may be practiced without these specific details. In other instances, well-known structures and devices are schematically shown in order to simplify the drawing.

(21) In addition, the terms used in the present disclosure, such as technical and scientific terms, have its own meanings and can be comprehended by those skilled in the art, unless the terms are additionally defined in the present disclosure. That is, the terms used in the following paragraphs should be read on the meaning commonly used in the related fields and will not be overly explained, unless the terms have a specific meaning in the present disclosure.

(22) Referring to FIGS. 1 to 3, FIG. 1 is a partially perspective view of an electronic device 1 according to a first embodiment of the disclosure, FIG. 2 is a partially exploded view of the electronic device 1 in FIG. 1, and FIG. 3 is another partially exploded view of the electronic device 1 in FIG. 1.

(23) In this embodiment, the electronic device 1 is, for example, a server. The electronic device 1 includes a casing assembly 10 and a plurality of electronic components 30. The casing assembly 10 includes a casing 11, a cage 12, a supporting component 13, an engagement component 14 and an operation assembly 15.

(24) Walls (not given reference numerals) of the casing 11 surround an accommodation space 111. The cage 12 is located in the accommodation space 111 of the casing 11 and pivotably disposed in the casing 11 so as to be pivotable between a closed position (shown in FIG. 1) and an opened position (shown in FIG. 9). Specifically, the cage 12 includes a frame body 121 and a side cover 122. The frame body 121 is pivotably disposed in the casing 11 via, for example, a plunger (not shown), such that the cage 12 is pivotable about a rotation axis P via the plunger. The frame body 121 has an installation space 1211 and an opening 1212 communicating with each other. The installation space 1211 is formed and surrounded by side plates (not given reference numerals) of the frame body 121 for accommodating the electronic components 30. The electronic components 30 can be removed from the installation space 1211 through the opening 1212 or placed into the

installation space **1211** through the opening **1212**. The side cover **122** is fixed to one of the side plates of the frame body **121**. The electronic components **30** are, for example, hard disk drives, but the disclosure is not limited thereto.

(25) Then, referring to FIGS. 2 to 4, FIG. 4 is a partially cross-sectional view of the electronic device **1** in FIG. 1.

(26) The supporting component **13** is, for example, a link assembly. One end of the supporting component **13** is pivotably disposed on the casing **11**, and another end of the supporting component **13** is slidably disposed on the side cover **122**. Specifically, the side cover **122** has a guide hole **1221**, and the guide hole **1221** is, for example, a through hole and has a first end **1222** and a second end **1223** located opposite to each other, and the first end **1222** is located closer to the rotation axis P of the cage **12** than the second end **1223**. In addition, the supporting component **13** includes a link **131**, a wheel **132** and a positioning component **133**. An end of the link **131** is pivotably disposed on the casing **11**, and the wheel **132** is rotatably disposed on another end of the link **131** via the positioning component **133**. The positioning component **133**, for example, serves as a shaft of the wheel **132** and is screwed into the link **131**, such that the wheel **132** is positioned on the link **131** and rotatable relative to the link **131**. The wheel **132** is partially and movably located in the guide hole **1221** of the side cover **122**, and the wheel **132** is movable between the first end **1222** and the second end **1223** of the guide hole **1221**. The wheel **132** can help the link **131** to be smoothly moved relative to the cage **12** during the rotation movement of the cage **12** relative to the casing **11**.

(27) Note that the guide hole **1221** of the side cover **122** is not restricted to being a through hole, and the wheel **132** and the positioning component **133** of the supporting component **13** are optional components; in some other embodiments, the guide hole of the side cover may not be a through hole but a groove, and the supporting component may be a slide block. In such a case, the cooperation of the guide hole and the slide block of the supporting component can also help the link to be moved smoothly relative to the cage.

(28) In this embodiment, the casing assembly **10** may further include a torsion spring **16**. Two opposite ends of the torsion spring **16** are respectively fixed to the casing **11** and the supporting component **13**. The torsion spring **16** is used to force the supporting component **13** to pivot relative to the casing **11** for moving the cage **12** from the closed position toward the opened position selectively.

(29) The engagement component **14** is pivotably disposed on the side cover **122** of the cage **12**. The engagement component **14**, for example, has a first end **141**, a second end **142** and a hook structure **143**. The first end **141** and the second end **142** of the engagement component **14** are respectively located two opposite sides of a rotation axis P1 of the engagement component **14**. The hook structure **143** is located at the first end **141** of the engagement component **14**. When the wheel **132** of the supporting component **13** is located at the second end **1223** of the guide hole **1221**, the hook structure **143** is engaged with the wheel **132** of the supporting component **13** so as to fix the cage **12** in the opened position. The hook structure **143** has an inclined guide surface **1431**. When the wheel **132** is moved from the first end **1222** toward the second end **1223** of the guide hole **1221**, the inclined guide surface **1431** is pushed by the wheel **132**, and the cage **12** is moved from the closed position toward the opened position.

(30) In this embodiment, the casing assembly **10** may, for example, include a first elastic component **17**. The first elastic component **17** is, for example, an extension spring. Two opposite ends of the first elastic component **17** are respectively fixed to the side cover **122** of the cage **12** and the second end **142** of the engagement component **14**. The first elastic component **17** applies a force to the engagement component **14** to force the hook structure **143** to be located in a movement path of the supporting component **13** (e.g., a path between the first end **1222** and the second end **1223** of the guide hole **1221**). Also, the first elastic component **17** can maintain the hook structure **143** to be engaged with the wheel **132** of the supporting component **13** when the wheel **132** of the

supporting component **13** is located at the second end **1223** of the guide hole **1221**.

(31) The operation assembly **15** includes a latch component **151** and a connection component **152**. The latch component **151** and the connection component **152** are slidably disposed on the side cover **122**, and a slidable direction **D1** of the latch component **151** is, for example, at an angle to a slidable direction **D2** of the connection component **152**. For example, the latch component **151** may have two first elongated through holes **1511**, and the connection component **152** may have two second elongated through holes **1521**. The casing assembly **10** may further include two first fasteners **18** and two second fasteners **19**. The two first fasteners **18** are, for example, screws. The first fasteners **18** are respectively disposed through the two first elongated through holes **1511** and are screwed into the side cover **122**, such that the latch component **151** is movable relative to the cage **12** along the direction **D1**. The second fasteners **19** are, for example, screws. The second fasteners **19** are respectively disposed through the second elongated through holes **1521** and are screwed into the side cover **122**, such that the latch component **151** is movable relative to the cage **12** along the direction **D2** that is, for example, perpendicular to the direction **D1**.

(32) Note that the slidable direction **D1** of the latch component **151** is not restricted to being perpendicular to the slidable direction **D2** of the connection component **152**; in some other embodiments, the slidable direction of the latch component and the slidable direction of the connection component may be at an angle other than a right angle.

(33) In this embodiment, the latch component **151** may have an inclined guide slot **1512**, and the connection component **152** may have a pushed protrusion **1522** and a pushing protrusion **1523**. The pushed protrusion **1522** is located in the inclined guide slot **1512**, and the pushing protrusion **1523** is in contact with a recess **1421** of the second end **142** of the engagement component **14**.

(34) In this embodiment, the operation assembly **15** may further include a slide knob **153**, and the frame body **121** of the cage **12** may further have a through hole **1213**. The slide knob **153** is disposed through the through hole **1213** and fixed to the latch component **151**. The slide knob **153** is slidable to move the latch component **151**. Note that the slide knob **153** may be an optional component; in some other embodiments, the operation assembly may not include the slide knob, and a protrusion structure may be provided to be integrally connected to the latch component and disposed through the through hole **1213** of the frame body **121** for being operated by a user.

(35) In addition, the casing **11** may have a positioning structure **112**, and the latch component **151** may have a positioning recess **1513**. The positioning structure **112** is, for example, a protrusion and located in the accommodation space **111** of the casing **11**. The positioning recess **1513** of the latch component **151** and the positioning structure **112** of the casing **11** are engaged with each other so as to keep the cage **12** to be in the closed position. In addition, the latch component **151** may further have an inclined contact surface **1514**. The inclined contact surface **1514** is located close to the positioning recess **1513** and located farther away from the slide knob **153** than the positioning recess **1513**. The inclined contact surface **1514** can be pushed by the positioning structure **112** of the casing **11** so as to move the latch component **151** away from the positioning structure **112**.

(36) In this embodiment, the casing assembly **10** may, for example, include a second elastic component **20**. The second elastic component **20** is, for example, an extension spring. Two opposite ends of the second elastic component **20** are respectively fixed to the side cover **122** of the cage **12** and the latch component **151**. The second elastic component **20** applies a force to the latch component **151** so as to keep the latch component **151** to be in a position where the positioning recess **1513** of the latch component **151** is engaged with the positioning structure **112** of the casing **11**.

(37) As shown in FIG. 4, when the cage **12** is in the closed position, the wheel **132** of the supporting component **13** is located at the first end **1222** of the guide hole **1221**, the inclined guide surface **1431** of the hook structure **143** of the engagement component **14** is located between the first end **1222** and the second end **1223** of the guide hole **1221** and is located closer to the second end **1223** of the guide hole **1221**, and the positioning recess **1513** of the latch component **151**

remains engaged with the positioning structure **112** of the casing **11** by the second elastic component **20**, such that the cage **12** is fixed in the closed position and is unable to be pivoted relative to the casing **11**. At this moment, the opening **1212** of the frame body **121** of the cage **12** is located in the accommodation space **111** of the casing **11**, and thus the user is unable to withdraw the electronic components **30** out of the cage **12** or place the electronic components **30** in the cage from the opening **1212** of the frame body **121** of the cage **12**.

(38) Then, referring to FIGS. 5 and 6, FIG. 5 is a partially cross-sectional view of the electronic device **1** in FIG. 4 when the slide knob **153** is operated, and FIG. 6 is a partially cross-sectional view of the electronic device **1** in FIG. 5 when the torsion spring **16** lifts up the cage **12**.

(39) When the electronic components **30** (shown in FIG. 2) are required to be withdrawn from the cage **12** or placed into the cage **12**, the slide knob **153** is firstly moved along a direction A to drive the positioning recess **1513** of the latch component **151** to disengage from the positioning structure **112** of the casing **11**. At this moment, the cage **12** is not fixed to the casing **11**, and thus the torsion spring **16** forces the supporting component **13** to pivot along a direction B so as to move the wheel **132** of the supporting component **13** toward the second end **1223** from the first end **1222** of the guide hole **1221** along a direction C by a distance and drive the cage **12** to pivot relative to the casing **11** about the rotation axis P by an angle (e.g., along a direction D), such that the cage **12** is lifted up relative to the casing **11**. As a result, the user can force the cage **12** to further lift up along the direction D for enabling the opening **1212** of the cage **12** to be completely moved out of the accommodation space **111** of the casing **11**. Note that the torsion spring **16** is an optional component; in some other embodiments, the casing assembly may include a compression spring, and two opposite ends of the compression spring are respectively in contact with the bottom of the casing and the bottom of the cage. In such a case, when the positioning recess of the latch component is disengaged from the positioning structure of the casing, the compression spring can automatically lift up the cage relative to the casing. Alternatively, in another embodiment, the casing assembly may not include the torsion spring or the compression spring, and when the positioning recess of the latch component is disengaged from the positioning structure of the casing, the cage can be manually lifted up relative to the casing.

(40) Then, referring to FIGS. 7 to 9, FIG. 7 is a partially cross-sectional view of the electronic device **1** in FIG. 6 when the cage **12** is pivoted relative to the casing **11**, FIG. 8 is a partially cross-sectional view of the electronic device **1** in FIG. 7 when the engagement component **14** is engaged with the supporting component **13**, and FIG. 9 is a partially perspective view of the electronic device **1** in FIG. 1 when the casing **11** is in the opened position.

(41) The user can manually pivot the entire cage **12** along the direction D by hand. During the movement of the cage **12** along the direction D, the wheel **132** of the supporting component **13** presses against the inclined guide surface **1431** of the hook structure **143** of the engagement component **14** so as to pivot the engagement component **14** along a direction E to leave the path where the wheel **132** is moved from the first end **1222** to the second end **1223** of the guide hole **1221** and stretch the first elastic component **17**. After the wheel **132** of the supporting component **13** passes through the hook structure **143** of the engagement component **14**, the first elastic component **17** forces the hook structure **143** to return its original position, such that the hook structure **143** of the engagement component **14** is engaged with the wheel **132** of the supporting component **13**. As a result, the engagement component **14** holds the cage **12** to be in the opened position. At this moment, the opening **1212** of the cage **12** are exposed outside the accommodation space **111** of the casing **11**, and thus the user can withdraw the electronic components **30** from the cage **12** or place the electronic components **30** into the cage **12** through the opening **1212** of the frame body **121** of the cage **12**.

(42) Then, referring to FIG. 10, FIG. 10 is a partially cross-sectional view of the electronic device **1** in FIG. 8 when the engagement component **14** is disengaged from the supporting component **13**.

(43) When the cage **12** is required to be pivoted back to the closed position (as shown in FIG. 4)

from the opened position, the engagement relationship between the hook structure **143** of the engagement component **14** and the wheel **132** of the supporting component **13** is required to be disengaged in advance. Specifically, the user can slide the slide knob **153** along the direction A to drive the latch component **151** to move along the direction A relative to the cage **12**, such that the latch component **151** moves the connection component **152** along a direction F via the inclined guide slot **1512** and the pushed protrusion **1522** so as to force the pushing protrusion **1523** of the connection component **152** to push the recess **1421** of the engagement component **14**, thereby driving the hook structure **143** of the engagement component **14** to disengage from the wheel **132** of the supporting component **13** along the direction E. As a result, the cage **12** is released from the opened position, and the user can press the cage **12** downwardly along a direction G so as to pivot the cage **12** to the closed position from the opened position. During the movement of the cage **12** from the opened position to the closed position, the inclined contact surface **1514** of the latch component **151** on the cage **12** presses against the positioning structure **112** of the casing **11** so as to move the latch component **151** along the direction A and stretch the second elastic component **20**. When the user keeps pressing the cage **12** downwardly along the direction G, the inclined contact surface **1514** of the latch component **151** passes through the positioning structure **112**, and the second elastic component **20** returns to its original position, such that the positioning recess **1513** of the latch component **151** and the positioning structure **112** of the casing **11** are engaged with each other so as to fix the cage **12** in the closed position.

(44) In this embodiment, one end of the supporting component **13** is pivotably disposed on the casing **11**, and another end of the supporting component **13** is slidably disposed on the cage **12**, the engagement component **14** is pivotably disposed on the cage **12** and can be engaged with the supporting component **13** when the cage **12** is in the opened position, and the operation assembly **15** is movably disposed on the cage **12** and can drive the engagement component **14** to disengage from the supporting component **13**, and thus in a case that there is no handle provided in the casing assembly **10**, the cage **12** can still be fixed in the opened position or released from the opened position, while the appearance of the electronic device **1** can be maintained.

(45) In this embodiment, the pushing protrusion **1523** of the connection component **152** is in contact with the recess **1421** of the second end **142** of the engagement component **14**, but the disclosure is not limited thereto; in some other embodiments, the pushing protrusion of the connection component may be in contact with the first end of the engagement component, and the inclined guide slot of the latch component may extend toward another direction, such that the moving latch component can force the connection component to move upwardly via the inclined guide slot and the pushed protrusion so as to drive the hook structure of the engagement component to disengage from the wheel of the supporting component.

(46) In this embodiment, one of the supporting component **13**, the engagement component **14**, one end of the first elastic component **17**, the latch component **151**, the connection component **152** and one end of the second elastic component **20** are all disposed on the side cover **122**, but the disclosure is not limited thereto; in some other embodiments, the cage may not include the side cover, and the aforementioned components are disposed on the frame body of the cage.

(47) In this embodiment, the latch component **151** is directly disengaged from the positioning structure **112** of the casing **11** by operating the slide knob **153** to slide the latch component **151**, and the engagement component **14** is disengaged from the supporting component **13** by operating the slide knob **153** to drive the latch component **151** to slide the connection component **152**, but the disclosure is not limited thereto.

(48) Referring to FIGS. **11** and **12**, FIG. **11** is a partially cross-sectional view of an electronic device **1a** according to a second embodiment of the disclosure, and FIG. **12** is a partially cross-sectional view of the electronic device **1a** in FIG. **11** when a button **153a** is pressed.

(49) The electronic device **1a** of this embodiment is similar to the electronic device **1** shown in FIGS. **1** to **10** of the previous embodiment, the main difference between them is the movement

relationship between the latch component and the connection component, and thus the following paragraphs will mainly introduce the difference between them, while other parts of this embodiment can be referred to the previous paragraphs related to the electronic device **1** and will not repeatedly introduced hereinafter.

(50) In this embodiment, an operation assembly **15a** does not include the slide knob, but include a button **153a**, and the button **153a** is fixed to a latch component **151a**. The latch component **151a** has an inclined guide slot **1512a**, and a connection component **152a** has a first pushing protrusion **1522a** and a second pushing protrusion **1523a**. The first pushing protrusion **1522a** is located in the inclined guide slot **1512a**, and the second pushing protrusion **1523a** is in contact with the recess **1421** of the second end **142** of the engagement component **14**.

(51) As shown in FIGS. **11** and **12**, in a case that the cage **12** is in the closed position, and the positioning recess **1513a** of the latch component **151a** is engaged with the positioning structure **112** of the casing **11**, the button **153a** can be pressed along a direction H to force the connection component **152a** to slide relative to the cage **12** along the direction F so as to drive the latch component **151a** to move along the direction A via the first pushing protrusion **1522a** and the inclined guide slot **1512a** for disengaging the positioning recess **1513a** of the latch component **151a** from the positioning structure **112** of the casing **11**. As a result, the cage **12** can be pivoted to the closed position from the opened position along the direction D.

(52) Then, referring to FIGS. **13** and **14**, FIG. **13** is a partially cross-sectional view of the electronic device **1** in FIG. **11** when the engagement component **14** is engaged with a supporting component **13**, and FIG. **14** is a partially cross-sectional view of the electronic device **1** in FIG. **13** when the engagement component **14** is disengaged from the supporting component **13**.

(53) As shown in FIGS. **13** and **14**, in a case that the cage **12** is in the opened position, and the engagement component **14** is engaged with the supporting component **13**, the button **153a** can be pressed along the direction H to force the connection component **152a** to slide relative to cage **12** along the direction F so as to pivot the engagement component **14** along the direction E via the second pushing protrusion **1523a** for disengaging the engagement component **14** from the supporting component **13**. As a result, the cage **12** can be pivoted to the closed position from the opened position.

(54) In this embodiment, the operation assembly **15a** includes the latch component **151a**, the connection component **152a** and the button **153a**, but the disclosure is not limited thereto; in some other embodiments, the operation assembly may not include the latch component, but include the connection component and the button, and the casing may not have the positioning structure. In such a case, the cage in the closed position can be fixed to the casing via other suitable components or structures. On the other hand, in another embodiment, the operation assembly may merely include the latch component and the slide knob. For example, referring to FIGS. **15** and **16**, FIG. **15** is a partially cross-sectional view of an electronic device **1** according to a third embodiment of the disclosure, and FIG. **16** is a partially cross-sectional view of the electronic device **1** in FIG. **15** when a slide knob **153b** is operated.

(55) The electronic device **1b** of this embodiment is similar to the electronic device **1** shown in FIGS. **1** to **10** of the previous embodiment, the main difference between them is the structure of the operation assembly, and thus the following paragraphs will mainly introduce the difference between them, while other parts of this embodiment can be referred to the previous paragraphs related to the electronic device **1** and will not repeatedly introduced hereinafter.

(56) In this embodiment, an operation assembly **15b** does not include the connection component, and a latch component **151b** of the operation assembly **15b** includes a main portion **1511b** and a pushing portion **1512b** connected to each other. The main portion **1511b** of the latch component **151b** has a positioning recess **15111b**, and the positioning recess **15111b** is used to be engaged with the positioning structure **112** of the casing **11**. The engagement component **14b** has a first inclined surface **144b**, and the pushing portion **1512b** has a second inclined surface **15121b**, and the second

inclined surface **15121b** is in contact with the first inclined surface **144b**. The slide knob **153b** is fixed to the main portion **1511b** of the latch component **151b**.

(57) As shown in FIGS. **15** and **16**, in a case that the cage **12** is in the closed position, and the positioning recess **15111b** of the latch component **151b** is engaged with the positioning structure **112** of the casing **11**, the slide knob **153b** can be moved along the direction A to move the latch component **151b** relative to the cage **12** for disengaging the positioning recess **15111b** of the main portion **1511b** of the latch component **151b** from the positioning structure **112** of the casing **11**. As a result, the cage **12** can be pivoted to the opened position to the closed position along the direction D.

(58) Then, referring to FIGS. **17** and **18**, FIG. **17** is a partially cross-sectional view of the electronic device **1** in FIG. **15** when the engagement component **14b** is engaged with the supporting component **13**, and FIG. **18** is a partially cross-sectional view of the electronic device **1** in FIG. **17** when the engagement component **14b** is disengaged from the supporting component **13**.

(59) As shown in FIGS. **17** and **18**, in a case that the cage **12** is in the opened position, and the engagement component **14b** is engaged with the supporting component **13**, the slide knob **153b** can be moved along the direction A to move the latch component **151b** relative to the cage **12** so as to force the pushing portion **1512b** to pivot the engagement component **14b** along the direction E via the first inclined surface **144b** and the second inclined surface **15121b** for disengaging the engagement component **14b** from the supporting component **13**.

(60) According to the casing assemblies and the electronic devices as discussed in the above embodiments, one end of the supporting component is pivotably disposed on the casing, and another end of the supporting component is slidably disposed on the cage, the engagement component is pivotably disposed on the cage and can be engaged with the supporting component when the cage is in the opened position, and the operation assembly is movably disposed on the cage and can drive the engagement component to disengage from the supporting component, and thus in a case that there is no handle provided in the casing assembly, the cage can still be fixed in the opened position or released from the opened position, while the appearance of the electronic device can be maintained.

(61) It will be apparent to those skilled in the art that various modifications and variations can be made to the present disclosure. It is intended that the specification and examples be considered as exemplary embodiments only, with a scope of the disclosure being indicated by the following claims and their equivalents.

Claims

1. A casing assembly, comprising: a casing; a cage, movably disposed in the casing; a supporting component, wherein one end of the supporting component is pivotably disposed on the casing, and another end of the supporting component is slidably disposed on the cage; an engagement component, pivotably disposed on the cage and selectively engaged with the supporting component to position the cage; and an operation assembly, movably disposed on the cage and connected to the engagement component, wherein the operation assembly is movable relative to the cage, and the engagement component is driven to disengage from the supporting component via the operation assembly selectively.
2. The casing assembly according to claim 1, wherein the operation assembly comprises a latch component and a connection component, the latch component and the connection component are slidably disposed on the cage, and the latch component is connected to the engagement component via the connection component.
3. The casing assembly according to claim 2, wherein a slidable direction of the latch component is different from a slidable direction of the connection component.
4. The casing assembly according to claim 2, wherein the casing has a positioning structure, and

- the latch component is selectively meshed with the positioning structure to position the cage.
5. The casing assembly according to claim 2, wherein the latch component has a inclined guide slot, the connection component has a pushed protrusion and a pushing protrusion, the pushed protrusion is disposed in the inclined guide slot, the pushing protrusion is in contact with the engagement component; the pushed protrusion of the connection component is selectively pushed by the inclined guide slot of the latch component, and the engagement component is selectively driven to disengage from the supporting component via the pushing protrusion of the connection component.
 6. The casing assembly according to claim 2, wherein the operation assembly further comprises a slide knob, and the slide knob is fixed to the latch component.
 7. The casing assembly according to claim 2, wherein the operation assembly further comprises a button, and the button is fixed to the connection component.
 8. The casing assembly according to claim 7, wherein the latch component has an inclined guide slot, the connection component has a first pushing protrusion and a second pushing protrusion, the first pushing protrusion is disposed in the inclined guide slot, the second pushing protrusion is in contact with the engagement component; the inclined guide slot of the latch component is selectively pushed by the button via the first pushing protrusion, and the engagement component is driven to disengage from the supporting component via the second pushing protrusion selectively.
 9. The casing assembly according to claim 1, wherein the operation assembly comprises a latch component, the latch component comprises a main portion and a pushing portion connected to each other, the engagement component has a first inclined surface, the pushing portion has a second inclined surface; the first inclined surface of the engagement component is selectively pushed by the second inclined surface of the pushing portion of the latch component, and the engagement component is driven to disengaged from the supporting component selectively.
 10. The casing assembly according to claim 9, wherein the operation assembly further comprises a slide knob, and the slide knob is fixed to the main portion of the latch component.
 11. The casing assembly according to claim 1, wherein the cage further has a guide hole, the supporting component comprises a link and a wheel, one end of the link is pivotably disposed on the casing, the wheel is rotatably disposed on another end of the link, and the wheel is partially and movably located in the guide hole.
 12. The casing assembly according to claim 1, further comprising a torsion spring, wherein two opposite ends of the torsion spring are respectively fixed to the casing and the supporting component.
 13. An electronic device, comprising: a casing assembly, comprising: a casing; a cage, movably disposed in the casing; a supporting component, wherein one end of the supporting component is pivotably disposed on the casing, and another end of the supporting component is slidably disposed on the cage; an engagement component, pivotably disposed on the cage and selectively engaged with the supporting component to position the cage; an operation assembly, movably disposed on the cage and connected to the engagement component, wherein the operation assembly is movable relative to the cage, and the engagement component is driven to disengage from the supporting component via the operation assembly selectively; and a plurality of electronic components, removably accommodated in the cage.
 14. The electronic device according to claim 13, wherein the operation assembly comprises a latch component and a connection component, the latch component and the connection component are slidably disposed on the cage, and the latch component is connected to the engagement component via the connection component.
 15. The electronic device according to claim 14, wherein a slidable direction of the latch component is different from a slidable direction of the connection component.
 16. The electronic device according to claim 14, wherein the casing has a positioning structure, and the latch component is selectively meshed with the positioning structure to position the cage.

17. The electronic device according to claim 14, wherein the operation assembly further comprises a slide knob, and the slide knob is fixed to the latch component.
 18. The electronic device according to claim 14, wherein the operation assembly further comprises a button, and the button is fixed to the connection component.
 19. The electronic device according to claim 13, wherein the operation assembly comprises a latch component, the latch component comprises a main portion and a pushing portion connected to each other, and the engagement component is selectively driven to disengage from the supporting component via the pushing portion of the latch component.
 20. The electronic device according to claim 19, wherein the operation assembly further comprises a slide knob, and the slide knob is fixed to the main portion of the latch component.
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